

Object-Oriented Programming (OOP, CCIT4023)

HKU SPACE Community College, 2022-2023

Assignment 3 (Written)

(15%)

(Total Marks: 60)

Instructions:

- Answer ALL questions in given “Answer Book” with HANDWRITING. Submit only the Answer Book.
 - Print out the given Answer Book for answering questions with HANDWRITING.
** In case students do NOT have printers, they may use A4-size blank/single-line paper(s) with format similar to the given answer book (e.g. draw proper section lines) to answer questions. Clearly indicate the question and section of each answer, with proper sectioning.*
 - Use black or dark-blue pen to answer. Make sure answers are well-written for assessment.
TYPING / POOR Writing may NOT be marked!!
 - Properly Scan/take-photo the finished answer book, convert it into ONE pdf file, and submit the pdf file to SOUL. Scanned/photo-taken copies (submitted to SOUL) must be clear enough for printing-out and marking. You may need to try in different light conditions while taking photos.
 - *** IMPORTANT *** Collusion and plagiarism are serious offences. Zero mark may be given, with possible further disciplinary action.
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- Finish this work, based on concepts and techniques learnt in our course. Submitted assessment work based on other non-taught concepts and techniques may be given zero mark or mark penalty.
 - Students should finish reviewing the related course notes and materials, before doing this assignment.
 - Individual work: Student MUST FINISH THIS WORK ALONE. Student cannot work with others.
 - * Plagiarism / Collusion / Shared work with others are not allowed. Zero mark will be given, with possible disciplinary action.*
 - Questions related to program codes are based on Java programming language, unless specified.
 - Follow given instructions and guidelines.
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Section A: (20 marks)

- *Identify the letter of the choice that best completes the statement or answers the question.*
- *Each question carries the same mark.*

1. What is the parameter type of the main() method, the entry point of Java program?
A) String
B) void
C) static
D) Array of String
2. Which of the following will not be inherited from a superclass in Java?
A) Fields
B) Constructors
C) Methods
D) None of the other answers
3. Which of the following will be matched last, for matching an identifier inside a method?
A) Field
B) Local variable
C) Parameter
D) None of the other answers
4. Given the statement below, which of the following is correct?

```
import aC.bC.cC;
```


A) aC.bC is a package.
B) import is a keyword.
C) cC is not a package.
D) All of the other answers
5. Which of the following modifiers should be used when declaring fields for internal usage only within the class itself?
A) protected
B) private
C) public
D) None of the other answers

6. Which of the following features supports more than one method sharing the same name in the same class?

- A) Java package
- B) Overriding
- C) Overloading
- D) Encapsulation

7. Given the valid statement below, which of the following is correct?

```
Circle[][] aCe;
```

- A) It creates an array of `Circle`.
- B) It declares an object of `Circle`.
- C) It declares an array of `Circle` array.
- D) None of the other answers

8. What is the value of the following expression, given the values of `abc` and `xyz` are 99 and 33?

```
(abc % xyz + "33")
```

- A) "0.033"
- B) "033"
- C) 033
- D) "0.33"

9. Which of the following declares an array?

- A) `String[] aS;`
- B) `double[] aD = new double[12];`
- C) `int[][][] ai;`
- D) All of the other answers

10. Given statement below, what is the value of `(doubleZ[1]+doubleZ.length)`?

```
String[] doubleZ = {"10", "2", "1024", "20"};
```

- A) 6
- B) 24
- C) 24.0
- D) None of the other answers

Section B: (40 marks)

- * Use concise and direct techniques / program codes we learn in our course. Useless or over-complicated techniques / program codes may be ignored or have penalty.
- * Ignore package statement, import statements or comments in program codes, unless specified.
- * Proper indentations required in writing program codes, unless specified.

1. (10 marks)

a) There are three major object-oriented features in object-oriented programming. Name them and briefly introduce each one of them in one sentence.

[6 marks]

b) Given the following expressions (with 5 integer variables **a, b, c, d, e**), identify the first and last operators to be evaluated in each expression.

- a < b | | c == d & & e > = a**
- a > = b & & c != d * e - b**

[4 marks]

2. (15 marks)

```
public class ProgLang { // represent Programming Language
    String pName; // represent its name
    protected int pCode; // represent its Code
    ProgLang(String inName, int inCode){
        pName = inName;
        pCode = inCode;
    }
}
```

Only consider the programme code given above, answer the following questions.

a) Identify and list all visibility modifiers that appear within the given code.

[2 marks]

b) Identify and list all primitive and reference types that appear within the given code.

[2 marks]

c) Draw the UML class diagram, only related to the given program code above.

[5 marks]

d) Write a new public method named `getInfo()` to this class, to get and return information.

- The method accepts no input parameter, but returns a string array.
 - The body creates and returns an array of two string elements, the first array element is the name and the second is the string converted from the code

[3 marks]

e) Write a new public method named `dispInfo()` to this class `ProgLang`, to display information of the object on console.

- The method accepts no input parameter, and has no return.
 - The body displays information on console in one line, similar to sample output below.

P Name: <name>; P Code: <code>

Sample Program Output

P Name: Python; P Code: 4020

[3 marks]

3. (15 marks)

Based on the class `ProgLang` in the last question 2 above, write a new Java class named `FunProgLang`, which is a subclass of this class. The new class includes:

- A class field which is a public constant, a *default CODE*, named `DEF_PCODE` having an integer value of `21xxx`, a number consisting 21 and the “last 3 digits” of your Student ID.
 - E.g. Use the number 21369, if your student ID is xxxxx369
- Two overloading constructors:
 - (2-arguments constructor) One constructor accepts two arguments: the first for setting the name field and the second for the code field. Its body simply calls its corresponding superclass constructor with these two input parameters.
 - (1-argument constructor) Another constructor accepts only one argument of string for setting the name. Its body simply calls its another constructor with the *default CODE* as its code, and the input parameter as its name.
- A main method, the entry point for starting the program. The method body
 - Creates an array containing 3 elements of `ProgLang` objects as in the table below. The last element is an object of `FunProgLang` and is created with the 1-argument constructor.
 - Uses a `for` loop to display the information of all array elements on console by calling the related display method, similar to the sample program output below.

Index	Name of Prog. Lang.	Code of Prog. Lang.	Type
0	Pytthon	4020	ProgLang
1	Java	4023	ProgLang
2	Quantum	Default	FunProgLang

Sample Program Output

P Name: Python; P Code: 4020
P Name: Java; P Code: 4023
P Name: Quantum; P Code: 21369

~ END ~